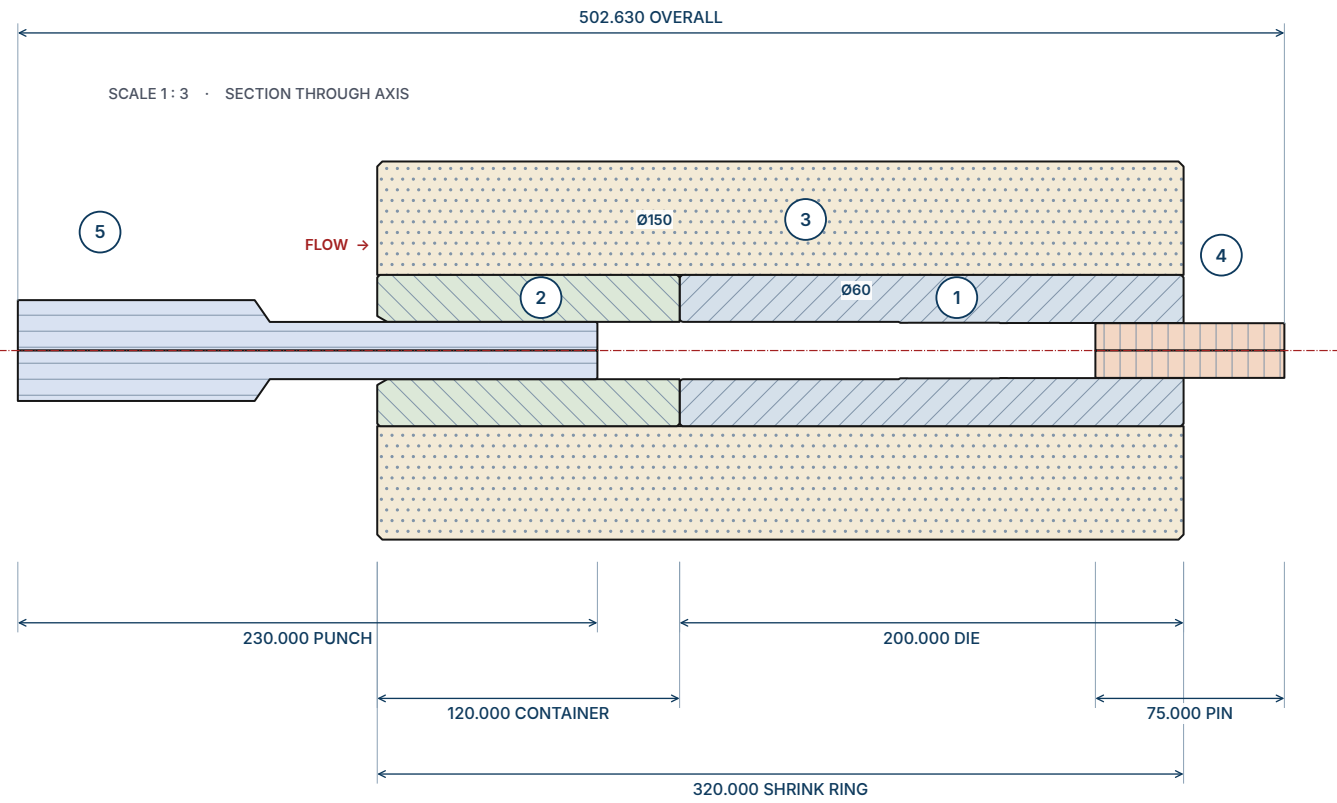




PARTS

	PART	MATERIAL	L	OD	ID
1	Die insert	M2/ASP23 62 HRC	200	60.000	22.750-21.860
2	Container	M2 62 HRC	120	60.000	22.750 / 27.350
3	Shrink ring	H13 46-48 HRC	320	150.000	60.000
4	Kickout pin	M2 60 HRC	75	21.700 / 21.100	solid
5	Punch	M2 60-62 HRC	230	40 / 22.700 / 21.700	solid

Principal diameters. Every station is on sheets 2-5.

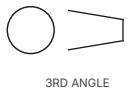
PIN, INSERT AND PUNCH — ID / OD / DIAMETRAL CLEARANCE (nominal, modelled assembly)

INTERFACE	MALE Ø	FEMALE Ø	Ø CLR.	RADIAL	WHERE IT RUNS
Pin body in die exit bore	21.700	21.860	0.160	0.080	Z -165.3 to -200
Pin nose in die exit bore	21.100	21.860	0.760	0.380	nose face at Z -165
Punch stem in container	22.700	22.750	0.050	0.025	Z +33.13 to +116
Punch nose in container	21.700	22.750	1.050	0.525	nose face at Z +32.63
Die OD in shrink ring	60.000	60.000	0.000	0.000	nominal, line to line
Container OD in shrink	60.000	60.000	0.000	0.000	nominal, line to line
Die guide / container bore	22.750	22.750	0.000	0.000	matched at datum A

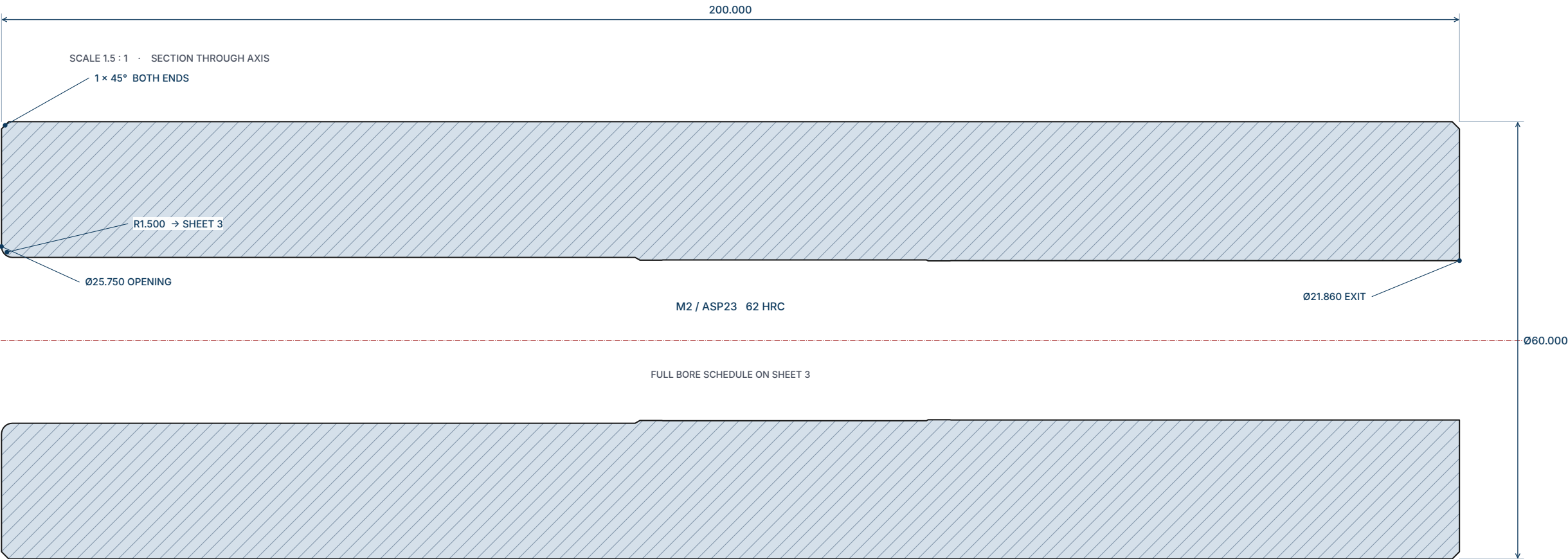
AXIAL STATIONS

STATION	STEP Z	FROM DATUM A
Punch head face	262.630	+262.630
Punch nose face	32.630	+32.630
Container mouth	120.000	+120.000
Datum A, entry face	0.000	0
Die exit face	-200.000	depth 200
Pin nose face	-165.000	depth 165
Pin back face	-240.000	depth 240

Pin insertion past the die exit face is 35.000 (nose) and the stickout is 40.000. The pin stays in the Ø21.860 exit bore. It does not enter the Ø21.800 land. Punch nose is 32.630 inside the container from datum A. Stem clearance on Ø22.750 is 0.050. At datum A the container face continues in to Ø22.750. The die opening there is Ø25.750, so the container leaves a 1.500 radial lip, equal to the entry radius, inside the die mouth. Both inserts carry a 1 × 45° OD chamfer, so the joint under the shrink ring is a 2 mm vee.

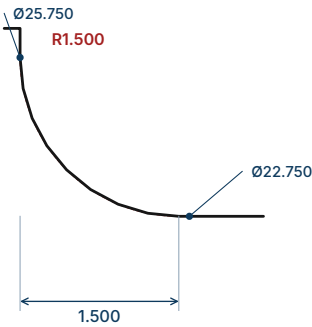


DRAWING	MCF000-OP1-R0	SHEET	1 / 5
PART	FORWARD EXTRUSION DIE	UNITS	MILLIMETRES
DATUM A	ENTRY FACE Z = 0	DATUM B	AXIS



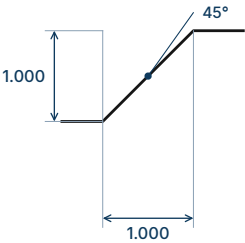
ENTRY BLEND 14 : 1

Dashed arc is the true R1.500. Solid line is the 8 facets.



OD CHAMFER 12 : 1 BOTH ENDS

Face Ø58.000 OD Ø60.000

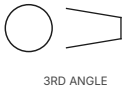


BORE, ENTRY TO EXIT

#	FEATURE	Ø FROM	Ø TO	LENGTH
1	Entry blend	25.750	22.750	1.500
2	Guide bore	22.750	22.750	85.42524
3	Reduction taper	22.750	22.000	0.64952
4	Bearing land	22.000	22.000	3.000
5	Relief taper	22.000	22.060	0.150
6	Relief bore	22.060	22.060	36.16266
7	Reduction taper	22.060	21.800	0.22517
8	Bearing land	21.800	21.800	3.000
9	Relief taper	21.800	21.860	0.150
10	Exit bore	21.860	21.860	69.73742

Angles, slant lengths and the eight facets are on sheet 3.

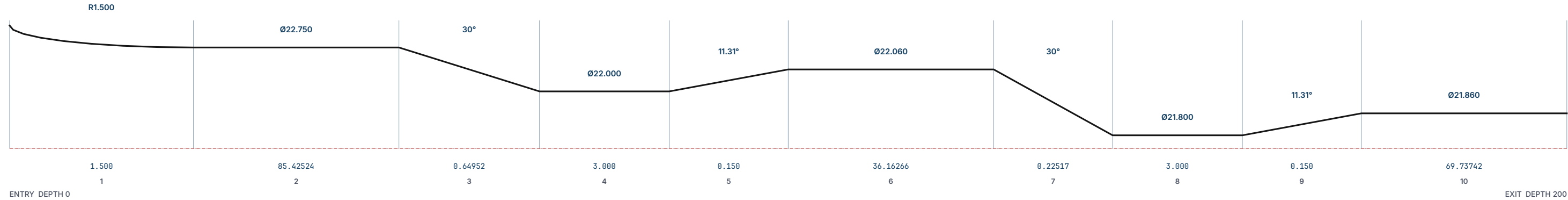
DEPTH INTO THE DIE IS -Z FROM DATUM A. FLOW IS LEFT TO RIGHT ON THIS SHEET.



DRAWING	MCF000-OP1-R0	SHEET	2 / 5
PART	FORWARD EXTRUSION DIE	UNITS	MILLIMETRES
DATUM A	ENTRY FACE Z = 0	DATUM B	AXIS

BORE MAP NOT TO SCALE · LONG PARALLELS SHORTENED · LENGTHS AND DIAMETERS ARE TRUE

Numbers under the map match the station table. Depth is measured from datum A toward the exit.



BORE STATIONS DEPTH = -Z DATUM A AT THE ENTRY FACE

#	FEATURE	DEPTH FROM	DEPTH TO	LENGTH	Ø FROM	Ø TO	ANGLE	SLANT	NOTE
1	Entry blend	0.000	1.500	1.500	25.750	22.750	R1.500	—	8 conical facets on R1.500
2	Guide bore	1.500	86.92524	85.42524	22.750	22.750	—	—	parallel
3	Reduction taper	86.92524	87.57476	0.64952	22.750	22.000	30°	0.750	ΔR 0.375 × √3
4	Bearing land	87.57476	90.57476	3.000	22.000	22.000	—	—	parallel
5	Relief taper	90.57476	90.72476	0.150	22.000	22.060	11.31°	0.15297	ΔØ +0.060 over 0.150, 1:5
6	Relief bore	90.72476	126.88742	36.16266	22.060	22.060	—	—	parallel, after the 0.060 relief
7	Reduction taper	126.88742	127.11258	0.22517	22.060	21.800	30°	0.260	ΔR 0.130 × √3
8	Bearing land	127.11258	130.11258	3.000	21.800	21.800	—	—	parallel
9	Relief taper	130.11258	130.26258	0.150	21.800	21.860	11.31°	0.15297	ΔØ +0.060 over 0.150, 1:5
10	Exit bore	130.26258	200.000	69.73742	21.860	21.860	—	—	kickout pin runs in this bore

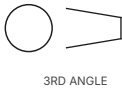
OUTSIDE DIAMETER

#	FEATURE	DEPTH FROM	DEPTH TO	LENGTH	Ø FROM	Ø TO	ANGLE
1	OD chamfer	0.000	1.000	1.000	58.000	60.000	45°
2	Outside diameter	1.000	199.000	198.000	60.000	60.000	—
3	OD chamfer	199.000	200.000	1.000	60.000	58.000	45°

ENTRY BLEND FACETS R1.500

STA	DEPTH	Ø	STEP	FACET ANGLE	STEP Z
1	0.000	25.750	0.02882	84.375°	0.000
2	0.02882	25.16473	0.08536	73.125°	-0.02882
3	0.11418	24.60195	0.13861	61.875°	-0.11418
4	0.25280	24.08329	0.18654	50.625°	-0.25280
5	0.43934	23.62868	0.22730	39.375°	-0.43934
6	0.66664	23.25559	0.25933	28.125°	-0.66664
7	0.92597	22.97836	0.28139	16.875°	-0.92597
8	1.20736	22.80764	0.29264	5.625°	-1.20736
9	1.500	22.750	—	—	-1.500

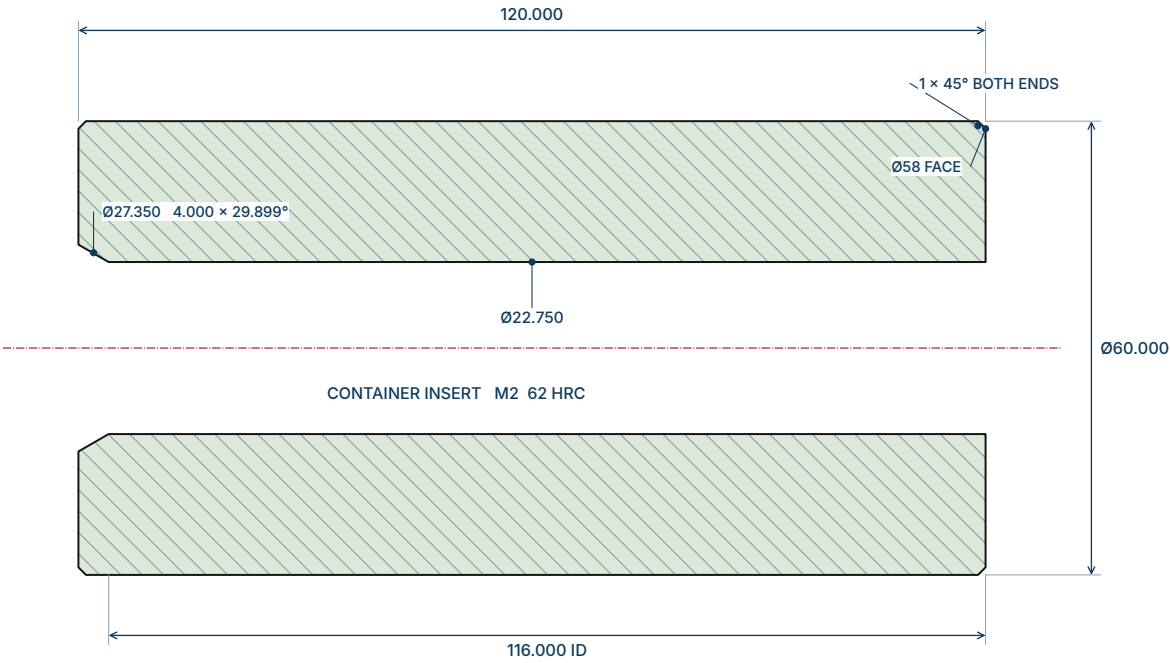
The nine facet stations lie on a circular arc of R1.500 centred at depth 1.500, radius 12.875. The arc is tangent to the entry face and to the Ø22.750 guide bore. Facet angles step by 11.250°. Each facet is a straight cone, so the surface is inside that arc by up to 0.00722 mm. Reduction angles are 30° from the axis (60° included). Axial length = ΔR × √3. Reliefs open the bore 0.060 on diameter over 0.150 axial (1:5, 11.310° from the axis).



ANGLE IS FROM THE AXIS. SLANT IS THE GENERATOR LENGTH. Ø CLR. OF THE PIN IS ON SHEET 1.

DRAWING	MCF000-OP1-R0	SHEET	3 / 5
PART	FORWARD EXTRUSION DIE	UNITS	MILLIMETRES
DATUM A	ENTRY FACE Z = 0	DATUM B	AXIS

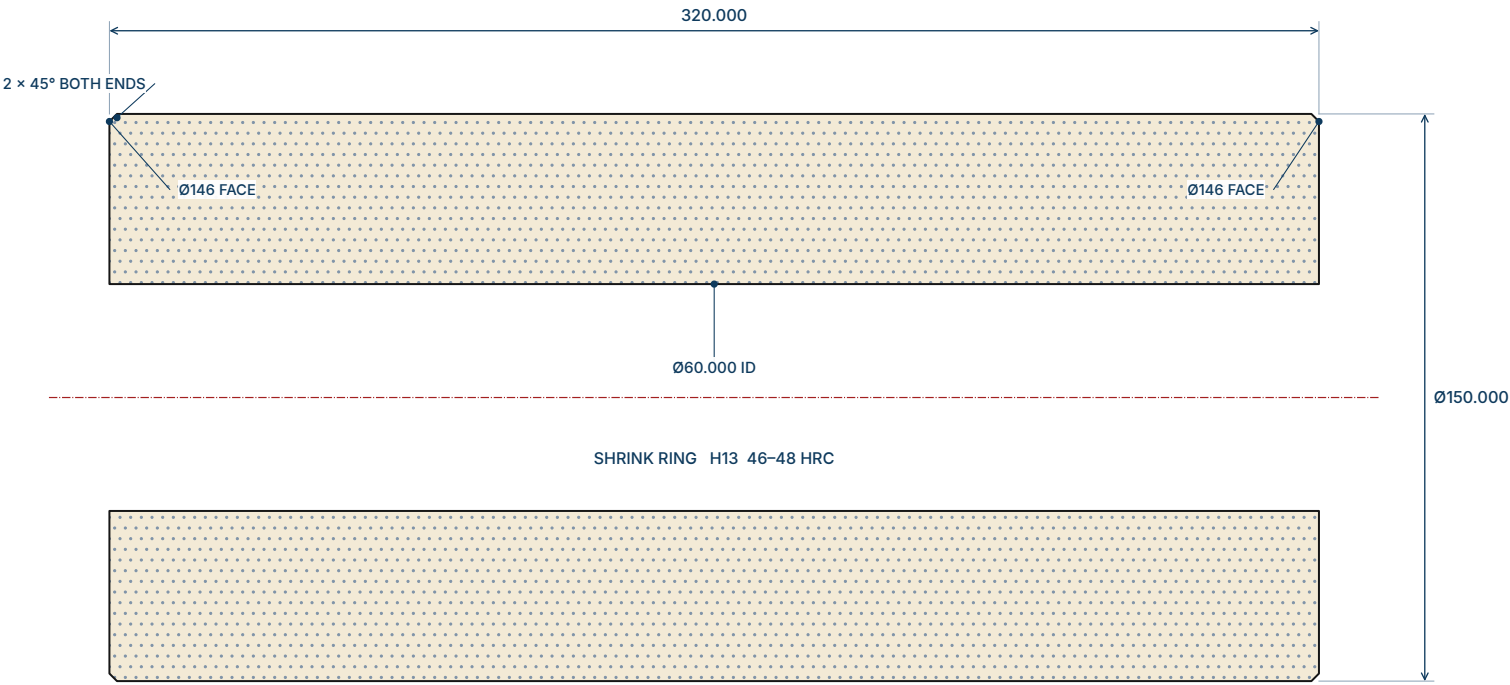
SCALE 1:1 · SECTION THROUGH AXIS



CONTAINER — ALL SEGMENTS

#	FEATURE	Z	LENGTH	Ø FROM	Ø TO	ANGLE
1	OD chamfer	120.000	1.000	58.000	60.000	45°
2	Face	120.000	0	27.350	58.000	—
3	ID taper	120.000	4.000	27.350	22.750	29.899°
4	OD	119.000	118.000	60.000	60.000	—
5	ID	116.000	116.000	22.750	22.750	—
6	OD chamfer	1.000	1.000	60.000	58.000	45°
7	Face	0.000	0	22.750	58.000	—

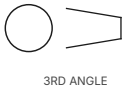
SCALE 1:2 · SECTION THROUGH AXIS



SHRINK RING — ALL SEGMENTS

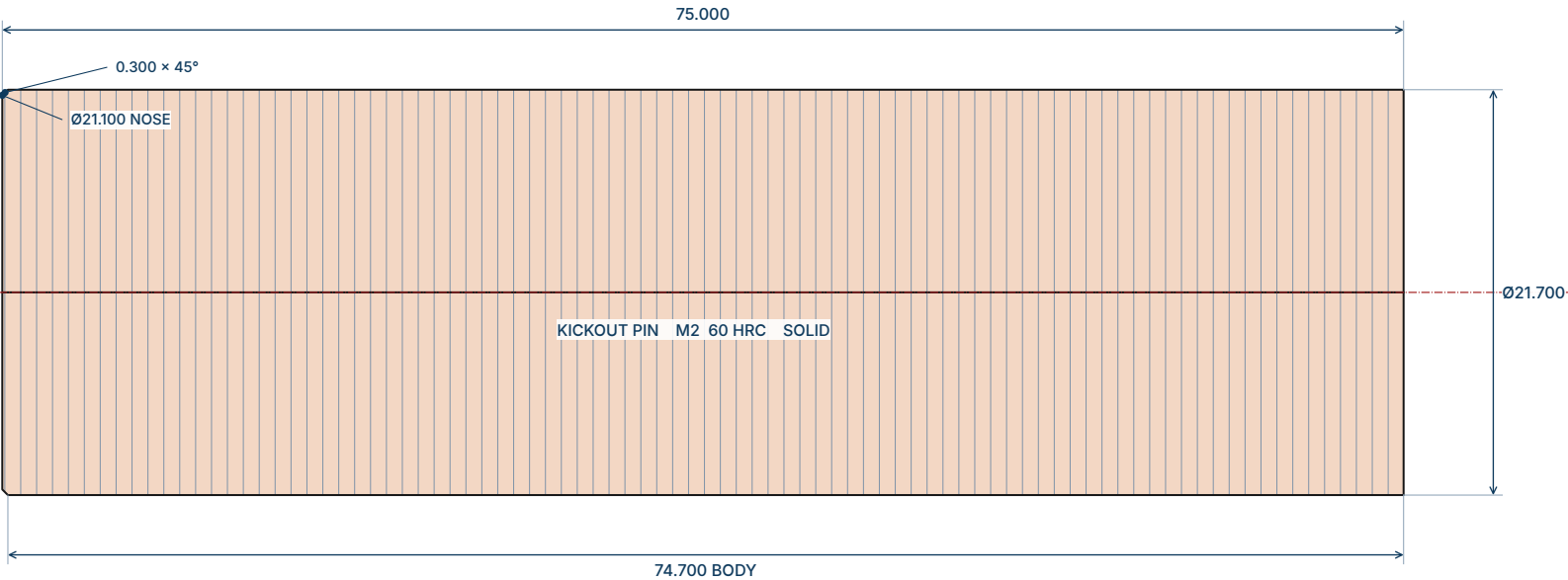
#	FEATURE	Z	LENGTH	Ø FROM	Ø TO	ANGLE
1	Face	120.000	0	60.000	146.000	—
2	OD chamfer	120.000	2.000	146.000	150.000	45°
3	ID	120.000	320.000	60.000	60.000	—
4	OD	118.000	316.000	150.000	150.000	—
5	OD chamfer	-198.000	2.000	150.000	146.000	45°
6	Face	-200.000	0	60.000	146.000	—

SHRINK RING IS FLUSH WITH THE CONTAINER MOUTH AND THE DIE EXIT. NOMINAL Ø60 IS LINE-TO-LINE WITH BOTH INSERTS.



DRAWING	MCF000-OP1-R0	SHEET	4 / 5
PART	FORWARD EXTRUSION DIE	UNITS	MILLIMETRES
DATUM A	ENTRY FACE Z = 0	DATUM B	AXIS

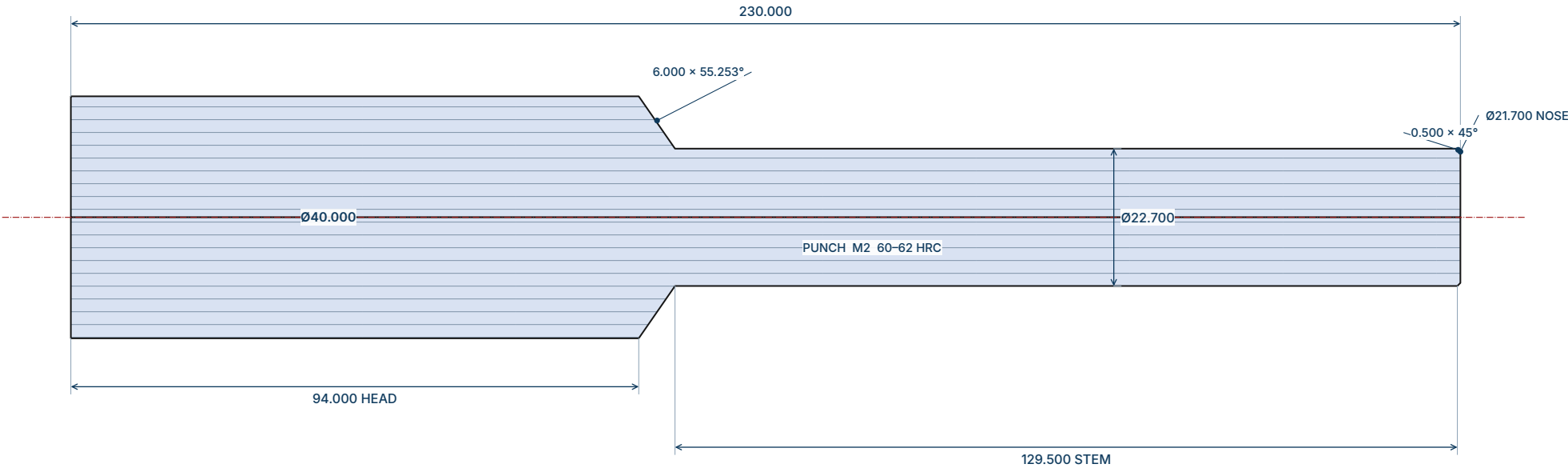
SCALE 2.5 : 1 · SECTION THROUGH AXIS



PIN AGAINST DIE EXIT BORE

Die exit bore ID	Ø21.860
Pin body OD	Ø21.700
Diametral clearance	0.160
Radial clearance	0.080
Nose face OD	Ø21.100
Nose diametral clr.	0.760
In the die / stickout	35.000 / 40.000
Does not enter Ø21.800 land	

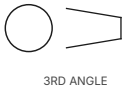
SCALE 1 : 1 · SECTION THROUGH AXIS



PUNCH AGAINST CONTAINER

Container ID	Ø22.750
Stem OD	Ø22.700
Stem Ø clearance	0.050
Radial	0.025
Nose OD	Ø21.700
Nose Ø clearance	1.050
Nose from datum A	32.630
Taper	6.000 long
Ø22.700 → Ø40.000	
55.253° from axis	
110.506° included	

PIN NOSE POINTS TOWARD THE DIE (LEFT ON THIS SHEET). PUNCH NOSE POINTS TOWARD THE DIE (RIGHT).



DRAWING	MCF000-OP1-R0	SHEET	5 / 5
PART	FORWARD EXTRUSION DIE	UNITS	MILLIMETRES
DATUM A ENTRY FACE Z = 0		DATUM B AXIS	